

Corpus Mind - Detailed Backend Setup Guide

For setting up the backend on a new local device

Overview

This guide assumes the user already has the backend source code.

Corpus Mind uses:

- FastAPI
- MongoDB Atlas (shared cluster)
- MongoDB Atlas Vector Search
- SentenceTransformers embeddings
- Ollama (llama3.1:8b)
- PyMuPDF / python-docx for document ingestion

Step 1 - Install Python

Install Python 3.11 or newer.

Verify installation:

```
python --version
```

Expected output should show Python 3.11+.

Step 2 - Install Git

Install Git and verify:

```
git --version
```

Required if cloning from GitHub.

Step 3 - Install Ollama

Install Ollama from <https://ollama.com/download>

Verify:

```
ollama --version
```

Corpus Mind uses Ollama for reasoning and answer generation.

Step 4 - Obtain Backend Files

Clone the repository:

```
git clone  
cd
```

Or download the project ZIP and extract it.

Step 5 - Create Virtual Environment

Windows:

```
python -m venv venv  
.\venv\Scripts\Activate.ps1
```

You should see (venv) in your terminal.

Step 6 - Install Python Dependencies

```
pip install -r requirements.txt
```

If requirements.txt is missing, install:

```
fastapi  
uvicorn  
pymongo  
python-dotenv  
sentence-transformers  
pymupdf  
python-docx  
httpx  
ollama
```

Step 7 - Configure Environment Variables

Create a .env file in the project root.

Example:

```
MONGO_URI=
```

Because this project uses a shared Atlas cluster, users do NOT need:

- Their own Atlas account
- Their own vector index
- Their own database

Step 8 - Download the LLM

Pull the model:

```
ollama pull llama3.1:8b
```

Verify:

```
ollama list
```

Expected:

```
llama3.1:8b
```

Step 9 - Start Ollama

Run:

```
ollama serve
```

Alternatively keep the Ollama desktop application running.
Without Ollama, the /ask endpoint will fail.

Step 10 - Start Backend

From the project root:

```
python -m uvicorn app.main:app --reload
```

Expected:

Application startup complete.
Uvicorn running on http://127.0.0.1:8000

Step 11 - Verify Swagger

Open:

<http://127.0.0.1:8000/docs>

Expected endpoints:

```
POST /ingest/file
POST /ingest/url
GET /search
POST /ask
GET /history
POST /conflicts
```

Step 12 - Upload a Document

Use POST /ingest/file.

Upload a PDF or DOCX.

Expected result:

```
{
  "doc_id": "...",
  "chunks_indexed": ...
}
```

Step 13 - Test Search

Use GET /search.

Example query:

tracking

Expected:

Results containing document chunks and similarity scores.

Step 14 - Test Question Answering

Use POST /ask.

Example:

```
{  
  "question": "Summarize this document"  
}
```

Expected:

Answer generated from retrieved chunks.

Troubleshooting

Error: Connection refused localhost:11434

→ Ollama is not running.

Error: model llama3.1:8b not found

→ Run:

ollama pull llama3.1:8b

Error: Could not connect to MongoDB

→ Check MONGO_URI in .env

Error: Frontend cannot connect

→ Ensure CORSMiddleware is enabled in FastAPI.

What New Users DO NOT Need To Do

Because the project uses a shared Atlas cluster:

They do NOT need to:

- Create MongoDB Atlas accounts
- Create databases
- Create collections
- Create Vector Search indexes
- Configure Atlas networking

The shared cluster already contains:

- corpus_mind database
- documents collection
- chunks collection
- history collection
- relations collection

- `chunk_vector_index` vector index